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Regional opportunities for CO_2 capture and storage in Southeast Asia

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ABSTRACT

In this paper, we evaluate opportunities for CO_2 capture and storage (CCS) within a study area of radius 1,000 km from Singapore. Results show that stationary CO_2 emission from the study area is 391 Mtpa. There is mid CO_2 storage resource of 0.5 Gt in oil reservoirs. In addition, CO_2 enhanced gas/oil recovery (CO_2-EGR/EOR) can recover 1.86 billion barrels of oil. The largest amount of CO_2-EOR is found in the Minas oil field (767 MMbbl) in Central Sumatra and the Seligi (119 MMbbl) and Dulang (99 MMbbl) oil fields in the Malay Basin. The mid CO_2 storage resource in gas reservoirs is 6.2 Gt. Of particular importance is the Arun gas condensate reservoir in the North Sumatra Basin with 1.3 Gt CO_2 storage resource and 101 MMbbl condensate recovery by CO_2-EGR. The mid CO_2 storage resource in saline aquifers is 379 Gt, accounting for the 98% of total CO_2 storage. Furthermore, we propose an ASEAN CCS corridor connecting Singapore with the North Sumatra, Central Sumatra, and Malay basins wherein CO_2 captured from Singapore and the host country can be transported to oil and gas reservoirs or saline aquifers within these basins for permanent storage.

1. Introduction

ASEAN is a regional grouping of ten nations in Southeast Asia, namely Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam. CO_2 emission from ASEAN has been trending upwards in the last five decades (Fig. 1), especially since the 1990s. As of 2019, CO_2 emission from the burning of fossil fuels and cement production in ASEAN was 1.67 Gt or 4.6% of total global CO_2 emission (Our World in Data, 2021). The biggest CO_2 emitters in ASEAN are Indonesia, Thailand, Malaysia, and Vietnam (Fig. 1), which are also the most populated countries.

All ten ASEAN nations are signatories to the Paris Agreement, and have pledged to reduce their greenhouse gas (GHG) emission voluntarily so that the earth's surface temperature rise is below 2°C from pre-industry times. However, to meet the aspiration of the Paris Agreement, ASEAN nations must peak their GHG emission by 2030 and achieve net-zero around 2050 (IPCC, 2019a; Lau et al., 2021a). This is a tall order , especially for many developing nations in ASEAN.

Many ways have been suggested for decarbonization including increased energy efficiency, electrification, renewable energies, nuclear energy, carbon dioxide utilization, etc. (DeAngelo et al., 2021; IPCC, 2014; IPCC, 2018. It is, however, generally agreed that CCS will be a necessary part of the solution to CO_2 emission reduction (Davis et al., 2018; IEA, 2021b; IPCC, 2019b; Pacala and Socolow, 2004). Without CCS, the decarbonization will be longer and more expensive (Lau, et al. 2021a; Metz et al., 2005). Consequently, there is an urgency to implement large-scale CCS projects globally. According to the Global CCS Institute (2020), there are currently 28 large-scale CCS projects operating worldwide, storing away 41 Mtpa. More are being planned. One such project is the Northern Lights Project in Norway which will capture 0.8 Mtpa of CO_2 from two industrial sources (a cement factory and a waste-to-energy plant) on the south coast of Norway (Equinor, 2019). The captured CO_2 will be shipped by liquid CO_2 tankers over approximately 1,000 km to Norway's west coast, from where it will be permanently stored in supercritical state in a saline aquifer at 2600 m below the sea bed. There are plans to expand the amount stored to 5 Mtpa by accepting CO_2 from other European countries.

There is increasing interest on CCS projects in ASEAN (IEA, 2021a). One CCS hub is proposed in East Java, Indonesia (ERIA, 2021). In addition, there is a proposal to ship CO_2 captured from SE Asia to Australia for storage (Zhang, 2020). However, from Singapore's perspective, East Java and especially Australia are rather far away for CO_2 storage. Locations of past and planned CCS pilots in the study area are shown in Fig. 2. Water-alternating-gas (containing more than 50% of CO_2) injection was applied in the Dulang oil field in the Malay Basin using six wells since November of 2002. Unfortunately, the pilot was suspended due to mechanical problems with the compressor (Kechut et al., 2001; Samsudin et al., 2005; Hamdan et al., 2005). There are plans for CO_2 storage pilots in the Bujang, Inas, and Sepat gas fields in the

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Nomenclature

CCS	Carbon capture and storage
CO_2-EOR	Carbon dioxide enhanced oil recovery
CO_2-EGR	Carbon dioxide enhanced gas recovery
EGR	Enhanced gas recovery
EOR	Enhanced oil recovery
Gt	10^9 tonne
OCIP	Original condensate-in-place, bbl
OGIP	Original gas-in-place, scf
OOIP	Original oil-in-place, bbl
Mt	10^6 tonne
Mtpa	Million ton per annum
scf	Standard cubic feet
MMbbl	Million barrels

ρ_{CO_2}	CO_2 density, kg/m^3
m_{CO_2}	Mass of the CO_2 storage, kg
V_g	Initial gas in place at the standard condition, Sm^3
B_g	Gas formation volume factor, m^3/Sm^3
R	Primary recovery factor, %
R_{CO_2}	Additional recovery factor by CO_2 injection, %
V_o	Original oil-in-place at standard conditions, Sm^3
B_o	Oil formation volume factor, m^3/Sm^3.
A	Aquifer area, m^2
h	Average net sand thickness, m
$\varnothing$	Average porosity of the target sand, %
E	CO_2 storage efficiency coefficient for oil and gas fields
E_{sa}	Efficiency factor for saline aquifer, %.Tcf 10^{12} scf

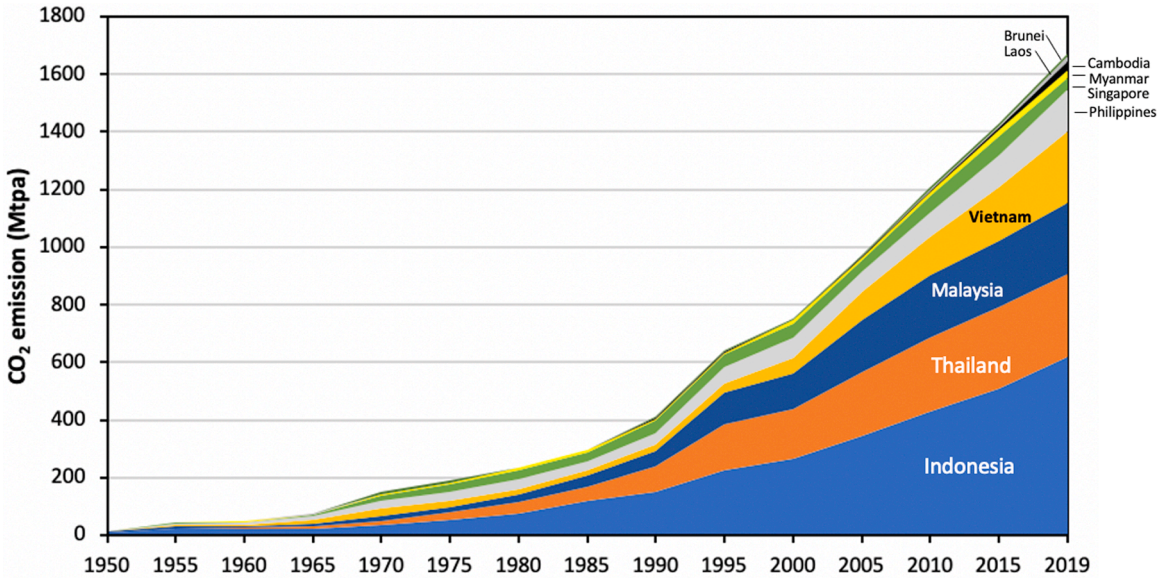


Fig. 1. CO_2 emission from fossil fuels and cement production in ASEAN (Our World in Data, 2021; World Bank Group, 2021).

Malay Basin in the next five years (Petronas, 2020). A geological carbon storage project in the Kaliberau gas field in South Sumatra is also planned (Koning et al., 2021).

In this paper, we explore the feasibility of a cross-border CCS project in ASEAN with Singapore as one of the sources of anthropogenic CO_2. We estimate stationary CO_2 sources and sinks within 1,000 km from Singapore. Stationary CO_2 sources are power plants, refineries, cement factories and steel and iron mills. Potential CO_2 sinks are oil and gas reservoirs and saline aquifers in Malaysia and Indonesia (Robinson, 1985; Murphy et al., 2005; Zahirovic et al., 2014; Purwanto et al. 2016; Doust, 2017). Within ASEAN, many oil reservoirs are in the mature development stage while most gas reservoirs are still in the early depletion stage (Riva, 1983; Seong and Hong, 1995; Todd et al., 1997; Bishop, 2002; Selamat et al., 2008; Hall, 2009; Yuwono et al., 2012; Iskandar et al., 2013). The study area consists of eight sedimentary basins (Table 1). They are the North Sumatra, Central Sumatra, South Sumatra, Northwest Java, Penyu, Malay, West Natuna and East Natuna basins (Fig. 3).

2. Method

In this study, stationary CO_2 emission in the study area is investigated plant-by-plant first. This is followed by field-by-field analyses of the CO_2 sinks including gas fields, oil fields and saline aquifers.

Most anthropogenic CO_2 emission comes from power plants, cement factories, iron and steel mills, and refineries (Metz et al., 2005). The amount of CO_2 emitted varies from plant to plant. The U.S. Energy Information Administration (EIA) compared CO_2 emission from power and industrial plants by fuel type. For power plants, the CO_2 emission estimate is based on the subbituminous coal power plant run time of 5,000 hours per year (Kishimoto et al., 2009; Evans et al., 2020; EIA, 2020a; EIA, 2020b). For cement production, 0.9 tonne of CO_2 is emitted per tonne of cement produced (Mahasenan et al., 2003; IndustryAbout, 2019; EIA, 2020a). For iron and steel mills, 1.9 tonnes of CO_2 are emitted for every tonne of steel produced (Jones, 2012; WSA, 2012; GEM, 2021). For refineries, we assume an average crude specific gravity of 0.86, an average refinery run time of 330 days per year with one month turnaround, and CO_2 emission of 95 lb per barrel of crude feed, according to EIA estimates on the performance of US refineries (Haynes,

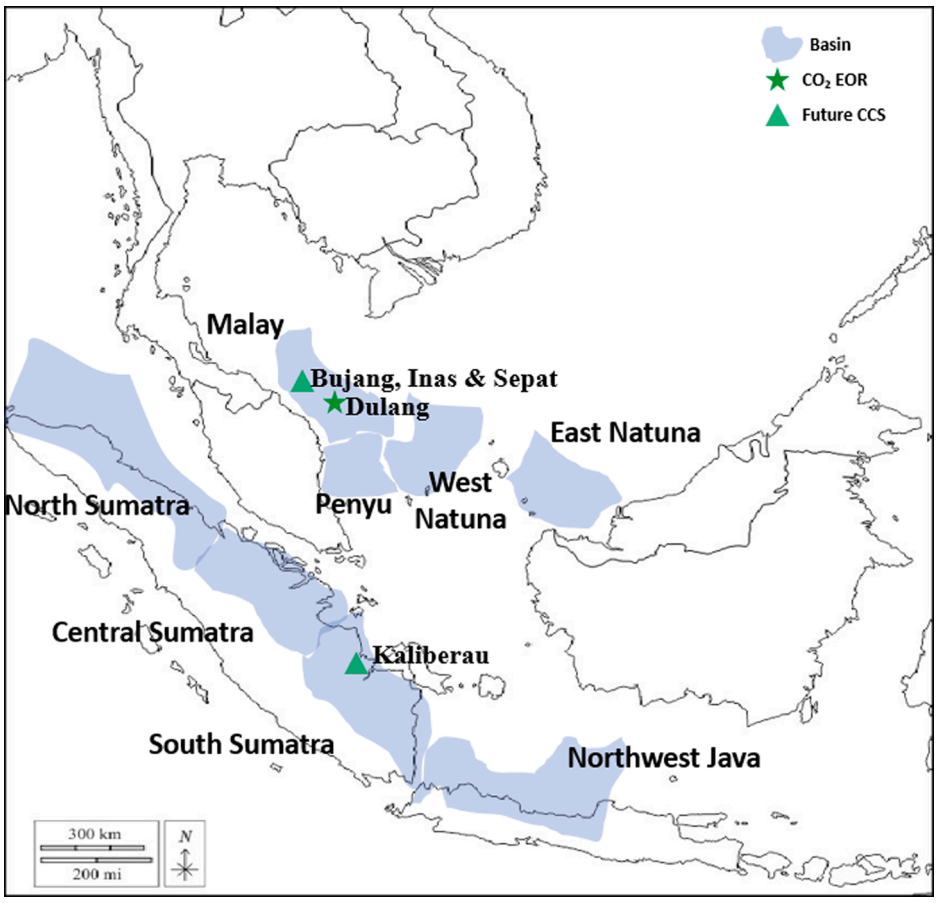


Fig. 2. CCS pilots in ASEAN.

Table 1
Oil and gas basins in the study area.

Basin	Distance from Singapore (km)
Central Sumatra	200
Penyu	230
South Sumatra	400
Malay	400
West Natuna	400
North Sumatra	800
Northwest Java	900
East Natuna	690

1976; Wang et al., 2004; Karras, 2011; FracTracker, 2021). This results in 0.043 tonne for CO_2 emission per barrel of crude feed to the refinery. The CO_2 emission estimates for different power and industry plants are summarized in Table 2.

CO_2 sinks include gas fields, oil fields and saline aquifers. Only major oil fields with original oil-in-place (OOIP) exceeding 100 MMbbl and gas fields with original gas-in-place (OGIP) exceeding 1 Tcf are investigated. The CO_2 storage estimations in oil and gas fields refer to the contingent resouces. The estimate of CO_2 storage in saline aquifers is based on prospective resources (SPE, 2017).

For a gas reservoir, the mass of CO_2 stored is given by (Bachu et al., 2007; Li et al, 2021):

$$m_{CO_2} = \rho_{CO_2} \times V_g \times B_g \times R \quad (1)$$

where m_{CO_2} is the mass of the CO_2 stored in kg; ρ_{CO_2} is the CO_2 density at reservoir conditions in kg/m^3; V_g is the initial gas-in-place at standard conditions in m^3; B_g is the gas formation volume factor in fraction; R is the primary recovery factor in percentage.

The CO_2 density can be calculated through the CO_2 properties calculator in the corresponding reservoir pressure and temperature (Megawatsoft, 2021). For a gas condensate reservoir, the condensate remaining in the reservoir is valuable and can be recovered through CO_2 injection. The mass of CO_2 stored is given by (APEC, 2005; Bachu et al., 2007):

$$m_{CO_2} = \rho_{CO_2} \times V_g \times B_g \times (R + R_{CO_2}) \quad (2)$$

where R is the recovery factor prior to CO_2 injection by primary and secondary recovery; R_{CO_2} is the additional recovery factor (%) by CO_2 injection.

For an oil reservoir, CO_2 can be used for enhanced oil recovery (EOR). The mass of CO_2 stored is given by (APEC, 2005; Bachu et al.,

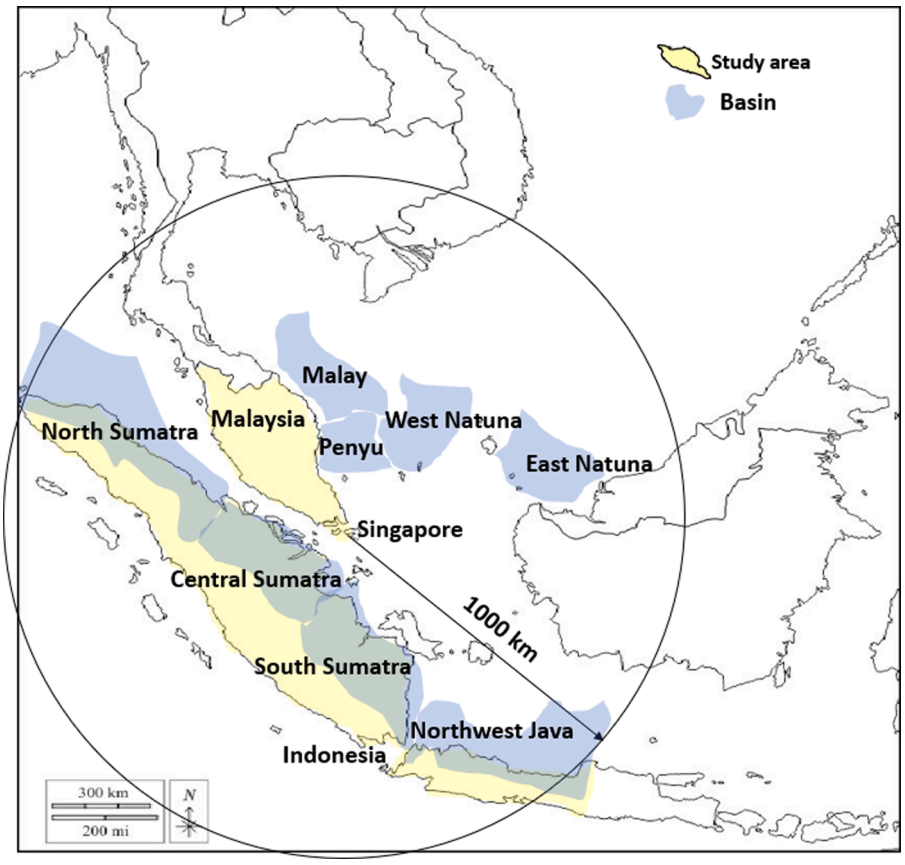


Fig. 3. Sedimentary basins in the study area.

Table 2
Estimates of CO_2 emission from power and industry plants in the study area

Plant type	Group	CO_2 emission	Unit	Reference
Power plant	Coal	1	tCO_2/MWh	(Evans et al., 2020)
	Natural gas	0.5	tCO_2/MWh	(IndustryAbout, 2019; EIA, 2020a; EIA, 2020b)
	Fuel oil	0.8	tCO_2/MWh	(IndustryAbout, 2019; EIA, 2020a; EIA, 2020b)
	Waste-to-energy	0.8	tCO_2/MWh	(Kishimoto et al., 2009; IndustryAbout, 2019; EIA, 2020a; EIA, 2020b)
Industry plant	Cement	0.9	tCO_2/t cement	(Mahasenan et al., 2003; IndustryAbout, 2019; EIA, 2020a)
	Iron and steel	1.9	tCO_2/t iron & steel	(Jones, 2012; WSA, 2012; IEA, 2020; GEM, 2021)
	Refinery	0.043	tCO_2/bbl crude	(Haynes, 1976; Wang et al., 2004; Karras, 2011; FracTracker, 2021)

2007; ADB, 2013):

$$m_{CO_2} = \rho_{CO_2} \times V_o \times B_o \times (R + R_{CO_2}) \quad (3)$$

where V_o is the initial oil-in-place at the standard conditions in m^3; B_o is the oil formation volume factor in rm^3/Sm^3; R is the recovery factor (%) by

Table 3
Recovery factor for primary and secondary recovery and CO_2-EOR.

Recovery mechanism	Recovery factor (%) Low	Mid	High	Reference
Primary gas recovery by pressure depletion	70	75	80	(Stoian and Telford, 1966; Sim et al., 2009)
Primary plus secondary oil recovery	30	35	40	(Zitha et al., 2011; Abedini et al., 2014; Li et al, 2021)
Immiscible CO_2-EOR	4.7	8.5	12.5	(Lake et al., 2014; Koottungal, L., 2014; Zhang et al., 2018; Farajzadeh et al., 2020; Li et al, 2021)
Miscible CO_2-EOR	8	12	15	(Lake et al., 2014; Koottungal, L., 2014; Qin et al., 2020; Li et al, 2021)

Table 4
CO_2 storage efficiency factor in a saline aquifer.

Reservoir type	CO_2 storage efficiency factor (%) Low	Mid	High	Reference
Aquifer	0.51	2	5.5	(Van der Meer, 1995; ADB, 2013; Bachu, 2015; DOE, 2015)

Table 5
CO_2 emission within the study area.

Basin/Region	Power plants (Mtpa)	Cement factories (Mtpa)	Iron & Steel mills (Mtpa)	Refineries (Mtpa)	Total (Mtpa)
Singapore	20.08	0	3.8	19.17	43.05
Peninsular Malaysia	74.74	23.36	31.33	7.11	136.54
North Sumatra	7.2	1.26	0	1.92	10.38
Central Sumatra	9.65	7.47	0	1.71	18.83
South Sumatra	10.39	1.80	28.5	0.71	41.4
NW Java	79.32	24.12	31.07	6.73	141.24
Subtotal	201.38	58.01	94.7	37.35	391.44

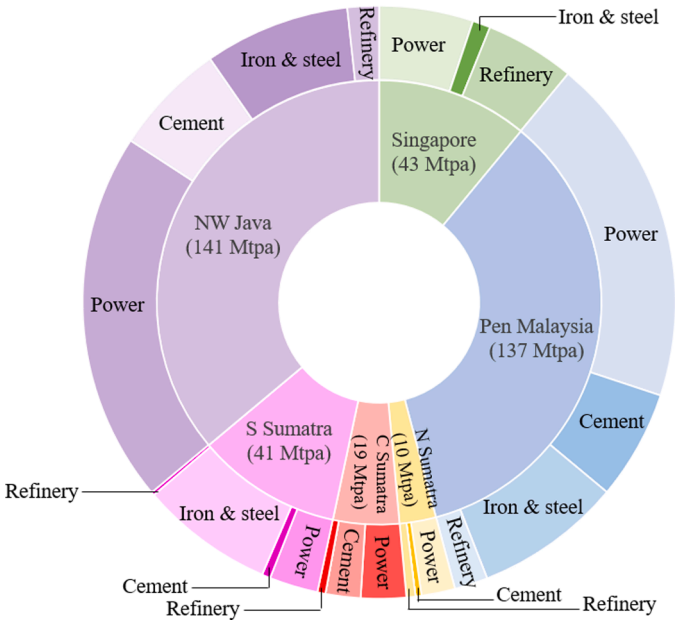


Fig. 4. CO_2 emission in the study area.

primary and secondary recovery; R_{CO2} the recovery factor (%) by CO_2 injection.

The CO_2-EOR recovery factor is related to the CO_2-oil miscibility. The minimum miscibility pressure (MMP) can be used to estimate the CO_2-oil miscibility. In the study, we use the following empirical correlation to calculate the CO_2 MMP (He et al., 2015; Li et al, 2021):

$$MMP = -329.558 + \left[7.727\left(\frac{8864.9}{API}\right)^{\frac{1}{1.012}}\left(1.005^{T}\right)\right] - 4.377\left(\left(\frac{8864.9}{API}\right)^{\frac{1}{1.012}}\right) \quad (4)$$

where T represents the reservoir temperature in °F.

For CO_2 storage in oil and gas fields, it is assumed that the pore space vacated by produced gas or oil is occupied by CO_2. Some researches assume that only part of this pore space will be occupied by CO_2. The rest may be occupied by injected water (Li et al., 2021). They, therefore, multiple m_{CO_2} by an additional CO_2 storage efficiency coefficient, E. The efficiency coefficient can be determined by reservoir simulations or local experiences. In this study we assume E to be unity for oil and gas fields in the absence of any field-specific data and detailed reservoir simulations.

For a saline aquifer, the mass of CO_2 stored is given by (APEC, 2005; Bachu et al., 2007):

$$m_{CO_2} = \rho_{CO_2} \times A \times h \times \varnothing \times E_{sa} \quad (5)$$

where A is the aquifer area in m^2, h is the average net sand thickness in m, $\varnothing$ is the average porosity of the target sand in %, E_{sa} is the efficiency factor in %.

Table 3 gives the low, mid, high values of the recovery factor of the gas depletion and CO_2-EOR taken from literatures. For oil reservoirs, we select only light oil with API of 27° and above to estimate the CO_2-EOR because medium to heavy oil reservoirs are not suitable for CO_2-EOR (Hamdan et al., 2005; Samsudin et al., 2005).

The CO_2 storage efficiency factor in saline aquifers has been studied and given in Table 4.

Having identified the CO_2 sources and sinks, a detailed source-sink mapping is performed to link one or more CO_2 sources to the nearest CO_2 sink. Finally, we consider the likely sinks where CO_2 from Singapore and a host country may be stored.

3. Results

The total CO_2 emission from the aforementioned sources are given in Table 5 and Figs. 4-5.

The locations for the CO_2 sources are plotted in the Fig. 5. Most stationary CO_2 emission comes from Singapore, Peninsular Malaysia and NW Java.

The CO_2 sinks in this study include only gas reservoirs with OGIP > 1 Tcf, oil reservoirs with OOIP > 100 MMbbl and saline aquifers located within 1,000 km from Singapore.

Properties of gas reservoirs in the study area are given in Tables 14 and 15 in Appendix A. Most of the gas reservoirs are still under primary depletion. For these gas reservoirs, CO_2 storage resource after the primary gas depletion is estimated using the primary recovery factor given in Table 3. The CO_2-EGR estimation is based on the miscible CO_2 recovery (Table 3). Among these gas condensate reservoirs, Arun (OGIP of 13 Tcf) is depleted and is the most valuable because large amount of condensate remains in the reservoir (OCIP of 840 MMbbl) (Table 14). Although the D-Alpha gas reservoir has an OGIP of 46 Tcf, it has not been developed (Table 14).

In a gas reservoir, we estimate the CO_2 storage resource based on voidage replacement (Klimkowski et al., 2015; Liu et al., 2021; Oldenburg et al., 2004). For a gas condensate reservoir, CO_2 injection leads to increased condensate production which also increases the amount of CO_2 stored. Results are presented in Tables 6–8. CO_2-EOR/EGR can be affected by many factors, such as well patterns, miscibility, CO_2 injection rate, oil price, carbon tax, etc (Liu et al., 2013; Melzer, 2012; Verma, 2015). The low, mid and high results are based on the P10, P50 and P90 recovery factors (Table 3). The mid CO_2 storage in the gas reservoirs within 1,000 km to Singapore is 6.2 Gt (Table 6). We compare our estimates of CO_2 storage resource in gas reservoirs with those by previous studies. Previous studies give CO_2 storage resource in gas reservoirs as 0.3 Gt in Central Sumatra, 0.8 Gt in the South Sumatra, 3.8 Gt in the Malay Basin, 0.64 Gt in the NW Java, 2.0 Gt in the North Sumatra (ADB, 2013; APEC, 2005). These estimates are bigger than those of our study. This discrepancy may be due to the following reasons. First, the previous studies include all the gas fields in each basin. We consider major gas fields with OGIP exceeding 1 Tcf only. Second, they assume a CO_2 density of 700 kg/m^3 and 100% recovery factor and use the total gas in place volume in each basin without consideration of gas reservoirs

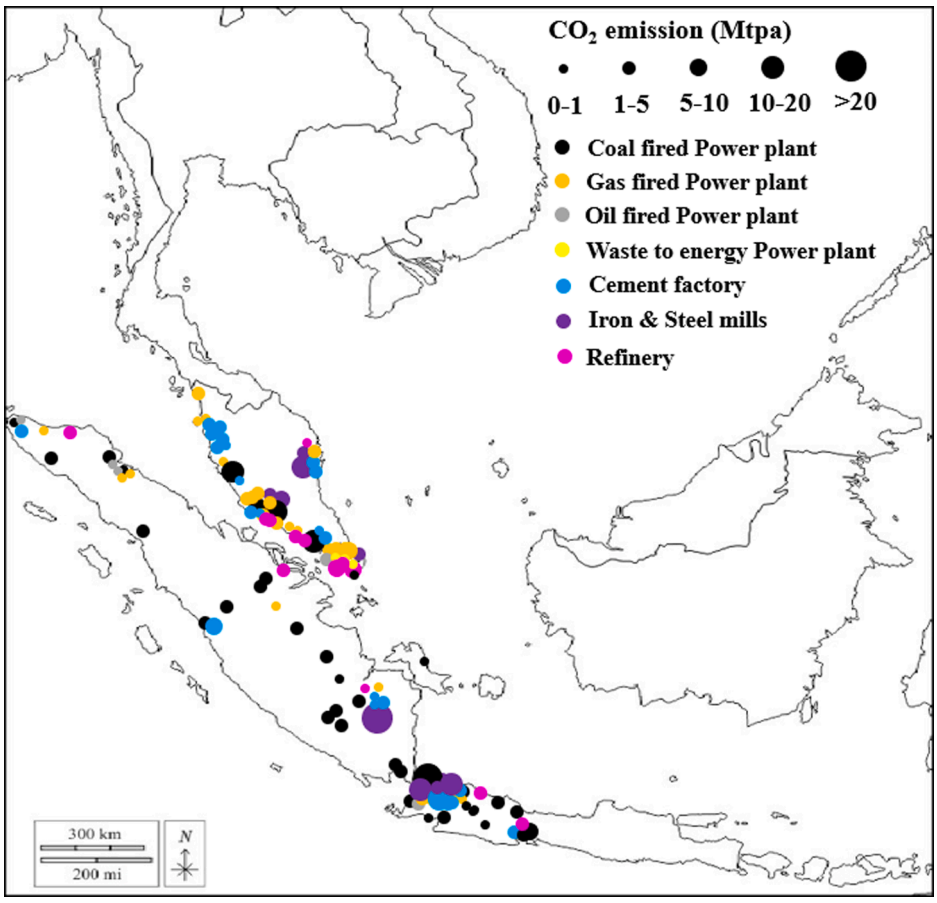


Fig. 5. CO_2 emission from the power and industry sectors within the study area.

Table 6
CO_2-EGR and CO_2 storage in gas reservoirs within the study area.

Country	Basin	CO_2 enhanced condensate recovery (MMbbl)			CO_2 storage (Mt)			CO_2 storage in previous studies (Mt)
		Low	Mid	High	Low	Mid	High	
Indonesia	Central Sumatra				64	68	73	300 (APEC, 2005; ADB, 2013)
	South Sumatra	27	40	50	559	599	639	800 (APEC, 2005; ADB, 2013)
	West Natuna	7	11	14	62	67	71	
	North Sumatra	67	101	126	1,717	1,792	1,840	2000 (APEC, 2005; ADB, 2013)
	NW Java				129	138	147	640 (APEC, 2005; ADB, 2013)
	East Natuna				2,026	2,171	2,316	
Malaysia	Malay	8	11	14	1,321	1,381	1,448	3800 (APEC, 2005; ADB, 2013)
Total		109	163	204	5,878	6,216	6,543	

Table 7
CO_2-EGR and CO_2 storage in Malaysian gas reservoirs within study area.

Basin	Gas Field	CO_2 enhanced condensate recovery (MMbbl)			CO_2 storage (Mt)		
		Low	Mid	High	Low	Mid	High
Malay	Duyong				139	149	159
	Jerneh				384	384	384
	Lawit				155	160	170
	Resak				78	84	89
	Bintang				107	115	122
	Tangga				79	85	91
	Bujang				75	80	86
	Inas				53	57	61
	Sepat				59	64	68
	Angsi	8	11	14	102	109	117
	Bergading				89	95	101
	Subtotal	8	11	14	1321	1381	1448

individually. We estimate the CO_2 density at the reservoir condition using the CO_2 properties calculator (Megawatsoft, 2021) and recovery factor field by field.

Besides that, additional condensate can be recovered by CO_2 injection. The mid CO_2 enhanced condensate recovery within 1,000 km from Singapore is 163 MMbbl (Table 6). Our calculations show that the condensate recovered by CO_2 injection is 101 MMbbl for the Arun reservoir, 40 MMbbl for Suban (Table 8), 11 MMbbl for Angsi (Table 7) and 11 MMbbl for Anoa (Tabale 8). Among them, Arun gas condensate reservoir is depleted and ready for CO_2 injection while the others are still under depletion.

Properties of oil reservoirs within the study area are listed Tables 16-17 in Appendix A. The earlier CO_2 is injected, the better is the CO_2-EOR performance. All light oil reservoirs in the study area can accept CO_2-EOR from today. However, CO_2-EOR is preferred in light oil reservoirs (API gravity $\geq 27°$) instead of medium to heavy oil (Hamdan et al., 2005; Samsudin et al., 2005). Oil reservoirs such as Duri in Central Sumatra,

Table 8
CO_2-EGR and CO_2 storage in Indonesian gas reservoirs within the study area.

Basin	Gas field	CO_2 enhanced condensate recovery (MMbbl)			CO_2 storage (Mt)		
		Low	Mid	High	Low	Mid	High
Central Sumatra	Bentu				64	68	73
South Sumatra	Suban	27	40	50	218	234	249
	Kaliberau				102	109	116
	Dayung				80	85	91
	Sumpal				95	102	109
	Musi				64	68	73
West Natuna	Anoa	7	11	14	62	67	71
North Sumatra	Arun	67	101	126	1,217	1,268	1,281
	Kuala Langsa				123	132	141
	S. Lhok Sukon				170	170	182
	Offshore NSB				207	221	236
NW Java	Parigi				129	138	147
East Natuna	D-Alpha				2,026	2,171	2,316
	Subtotal	101	152	190	4,557	4,833	5,085

and Bima, Sundari, Nurbani in NW Java are not suitable for CO_2-EOR because of their low oil gravity (Table 17). Consequently, CO_2 storage resource estimation for these reservoirs does not include additional recovery factors from CO_2-EOR. The biggest oilfield in Central Sumatra is Minas with an OOIP of 9 billion barrels (Table 17). Currently, the reservoir still contains 4.5 billion barrels of remaining oil. CO_2-EOR has the potential to reduce carbon footprints from the power and industry sectors. 81% of the global operational CCS projects are focused on CO_2-EOR (Global CCS Institute, 2020). CO_2-EOR delivers a lower-cost opportunity to deploy CCS projects (Cooney et al., 2015; Mavar et al., 2021).

Results of CO_2 storage resource estimate in oil reservoirs are given in Tables 9–11. The mid CO_2 storage resource in oil reservoirs is 0.5 Gt (Table 9). Oil reservoirs in the East Natuna Basin have in-place-volume less than 100 MMbbl and are not included in our study. We compare results of CO_2 storage resource in oil reservoirs with those by previous studies. Previous studies give CO_2 storage resource in the oil reservoirs at 1.7 Gt in Central Sumatra, 0.3 Gt in South Sumatra, 0.5 Gt in the Malay Basin, 0.4 Gt in the NW Java, 0.1 Gt in the North Sumatra (ADB, 2013; APEC, 2005), which are bigger than those in our study. This might be due to the following reasons. First, they estimate all the oil fields in the corresponding basins. We only consider the oil fields (OOIP) exceeding 100 MMbbl in our study. Second, their assume a CO_2 density of 700 kg/m^3 and 100% recovery factor using the total oil in-place volume in each basin without consideration of oil reservoirs individually. We estimate the CO_2 density at the reservoir condition and recovery factor field-by-field. Besides that, large amount of oil can be recovered by CO_2 injection. The amount of CO_2-EOR with mid case scenario is 1.9 billion barrels of oil (Table 9).

The locations of gas, oil reservoirs and aquifers are shown in Fig. 6.

Properties of saline aquifers within the study area are given in Table 12 in Appendix A. On average, a net-to-gross ratio of 0.3 is

Table 9
CO_2-EOR and CO_2 storage in oil reservoirs within the study area.

Country	Basin	CO_2 EOR (MMbbl)			CO_2 storage (Mt)			CO_2 storage in previous studies (Mt)
		Low	Mid	High	Low	Mid	High	
Malaysia	Penyu	10.9	19.7	29.0	7.4	8.1	9.6	
	Malay	265.0	479.3	704.9	165.5	186.7	217.4	500 (APEC, 2005; ADB, 2013)
Indonesia	Central Sumatra	545.4	986.4	1,450.6	147.0	160.3	174.4	1,700 (APEC, 2005; ADB, 2013)
	South Sumatra	60.0	108.5	159.5	44.4	48.4	53.0	300 (APEC, 2005; ADB, 2013)
	West Natuna	15.5	28.1	41.3	21.6	22.8	24.2	
	North Sumatra	24.0	36.0	45.0	7.0	7.5	8.1	100 (APEC, 2005; ADB, 2013)
	NW Java	110.1	199.1	292.8	58.4	66.2	75.1	400 (APEC, 2005; ADB, 2013)
Total		1,031	1,857	2,723	451	500	562	

Table 10
CO_2-EOR and CO_2 storage in Malaysian oil reservoirs within the study area.

Basin	Oilfield	CO_2 oil miscibility	CO_2 EOR (MMbbl)			CO_2 storage (Mt)		
			Low	Mid	High	Low	Mid	High
Penyu	Rhu-Ara	Immiscible	10.9	19.7	29.0	7.4	8.1	9.6
Malay	Guntong	Immiscible	47.0	85.0	125.0	31.9	34.9	41.2
	Seligi	Immiscible	65.8	119.0	175.0	44.3	48.6	58.6
	Tapis	Immiscible	47.0	85.0	125.0	31.5	34.2	37.0
	Pulai	Immiscible	6.1	11.0	16.1	3.0	3.2	3.7
	Bekok	Immiscible	22.4	40.5	59.5	18.0	19.6	21.2
	Sotong	Immiscible	10.5	19.0	28.0	6.6	8.4	10.1
	Belumut	Immiscible	11.4	20.7	30.4	4.2	5.3	6.3
	Dulang	Immiscible	54.8	99.2	145.8	26.0	32.5	39.3
	Subtotal		275.9	499.1	733.8	172.9	194.8	227

Table 11
CO_2-EOR and CO_2 storage in Indonesian oil reservoirs within the study area.

Basin	Oilfield	CO_2 oil miscibility	CO_2 EOR (MMbbl)			CO_2 storage (Mt)		
			Low	Mid	High	Low	Mid	High
Central Sumatra	Minas	Immiscible	424.1	767.0	1,127.9	105.2	112.4	120.0
	Duri	Not suitable for CO_2-EOR	0.0	0.0	0.0	6.3	7.3	8.4
	Bangko	Immiscible	30.4	55.0	80.9	5.3	5.9	6.4
	Bekasap	Immiscible	29.8	53.8	79.1	11.6	12.4	13.3
	Menggala	Immiscible	7.5	13.6	20.0	2.1	2.3	2.8
	Balam	Immiscible	13.2	23.8	35.0	2.1	2.6	3.1
	Petapahan	Immiscible	6.1	11.1	16.3	3.3	3.5	3.7
	Kotabatak	Immiscible	34.4	62.2	91.5	11.1	13.9	16.7
South Sumatra	Kuang	Immiscible	29.4	53.2	78.3	21.9	23.3	24.8
	Kaji-Semoga	Immiscible	18.8	34.0	50.0	12.9	14.0	15.1
	Sopa	Immiscible	4.7	8.5	12.5	3.4	4.3	5.2
	Jene	Immiscible	7.1	12.8	18.8	6.3	6.8	7.9
West Natuna	Udang	Immiscible	15.5	28.1	41.3	21.6	22.8	24.2
North Sumatra	Rantau	Miscible	24.0	36.0	45.0	7.0	7.5	8.1
NW Java	Cinta-rama	Immiscible	32.8	59.2	87.1	8.4	10.5	12.7
	Krisna	Immiscible	5.6	10.2	15.0	3.3	3.5	3.8
	Widuri	Immiscible	12.2	22.1	32.5	6.8	8.5	10.3
	Bima	Not suitable for CO_2-EOR	0.0	0.0	0.0	5.8	6.8	7.8
	Farida-Zelda	Immiscible	10.2	18.4	27.0	8.6	9.1	9.8
	Sundari	Not suitable for CO_2-EOR	0.0	0.0	0.0	1.3	1.5	1.7
	Nurbani	Not suitable for CO_2-EOR	0.0	0.0	0.0	0.5	0.6	0.7
	Ardjuna	Immiscible	32.1	58.1	85.4	14.1	15.1	16.1
	Arimbi	Immiscible	5.5	9.9	14.5	1.7	1.8	2.0
	Jatibarang	Immiscible	11.8	21.3	31.3	8.0	8.8	10.3
Total			755.2	1,358.3	1,989.4	278.6	305.2	334.9

assumed (Zhou et al., 2013) to arrive at the net sand thickness in our study. The CO_2 injection depth is assumed 1,000 m or deeper for CO_2 to be supercritical and the overburden forms a competent seal to prevent CO_2 leakage (APEC, 2005).

Results of CO_2 storage resource estimates are given in Table 12. The mid CO_2 storage resource in saline aquifers is 379 Gt. We compare our results of CO_2 storage resource with those by other studies. A previous study gave CO_2 storage resource in saline aquifers at 84 Gt in the Malay Basin by assuming a geothermal gradient of 25°C/km (Junin et al., 2016). The geothermal gradient used in the study area is above 50°C/km. Another previous study (ADB, 2013) gave CO_2 storage resource of 7.7 Gt in the South Sumatra Basin with 1% efficiency factor which is consistent with the low CO_2 storage resource estimate in our study (Table 12). Yet another previous study (Hedriana, 2017) gave CO_2 storage resource of 7.1 Gt in the NW Java Basin with 1% efficiency factor which is consistent with the low CO_2 storage resource estimate in our study (Table 12).

After the identification of stationary CO_2 emission and CO_2 sinks, a CO_2 source-sink mapping is conducted based on the least CO_2 transportation distance. The locations of stationary CO_2 sources and CO_2 sinks within the study area are plotted in Fig. 7. Their CO_2 emission and storage resource are summarized in Table 13. Total stationary CO_2 emission from power plants, steel and iron mills, cement factories and refineries within the study area is 391 Mtpa. Of this, 51% comes from power plants, 24% comes from iron and steel mills, 15% from cement factories, 10% from refineries. Histograms of stationary CO_2 emission and CO_2 storage are given in Fig. 8. The mean CO_2 emission for 128 power and industrial plants is 3 Mtpa (Fig. 8a). The mean CO_2 storage resource of 57 oil and gas fields in the study area is 122 Mt (Fig. 8b). The mean CO_2 storage resource of saline aquifers in the eight basins is 47 Gt (Fig. 8c).

There is 386 Gt of mid CO_2 storage resource in the study area which is enough for 987 years of CO_2 storage (Fig. 9 and Table 13). The most significant CO_2 storage resource is in saline aquifers of each sedimentary basin (Fig. 10). Overall, 98.3% (379 Gt) of storage resource comes from saline aquifers, 1.6% (6.2 Gt) from gas reservoirs, and 0.1% (0.5 Gt) from oil reservoirs (Fig. 10). The biggest CO_2 storage resource is found in the North Sumatra (113.8 Gt), Malay (77.3 Gt), East Natuna (62.1 Gt), NW Java (28.0 Gt), Penyu (27.7 Gt), and South Sumatra (27.6 Gt) basins (Fig. 9b).

In the North Sumatra Basin, the depleted Arun gas condensate field is the best candidate for CO_2-EGR and CO_2 storage (Table 8). It has 1.3 Gt of mid CO_2 storage resource and 101 MMbbl of mid condensate recovery by CO_2-EGR. In the Central Sumatra Basin, the Minas oilfield is the best candidate for CO_2-EOR and CO_2 storage. It has mid CO_2 storage resource of 112 Mt and mid CO_2-EOR recovery of 767 MMbbl (Table 11). In the South Sumatra Basin, the Suban gas condensate field, with a current recovery factor of 50%, will be a good candidate for CO_2-EGR and CO_2 storage after primary depletion. It has mid CO_2-EGR recovery of 40 MMbbl and mid CO_2 storage resource of 234 Mt (Table 8). In the NW Java Basin, light oil reservoirs are good candidates for CO_2-EOR and CO_2 storage. Gas reservoirs in this basin are still under depletion and not ready for CO_2 storage. The gas fields in the West Natuna Basin are still under depletion and not ready for CO_2 storage. The oil fields have limited resource for CO_2 storage. In the East Natuna Basin, the D-Alpha gas field is not yet developed and hence not ready for CO_2 storage (Table 14). In the Penyu Basin, the oil and gas fields have limited CO_2 storage resource. In the Malay Basin, the gas and oil reservoirs are good candidates for CO_2-EGR/EOR and CO_2 storage. They have a mid CO_2-EGR/EOR recovery of 490 MMbbl (Tables 7 and 10) In addition, the Jerneh gas field with a current recovery factor of 80% is a good candidate for CO_2 storage. It has a mid CO_2 storage resource of 384 Mt (Table 7).

Our analysis shows that there are enough CO_2 storage resource in the

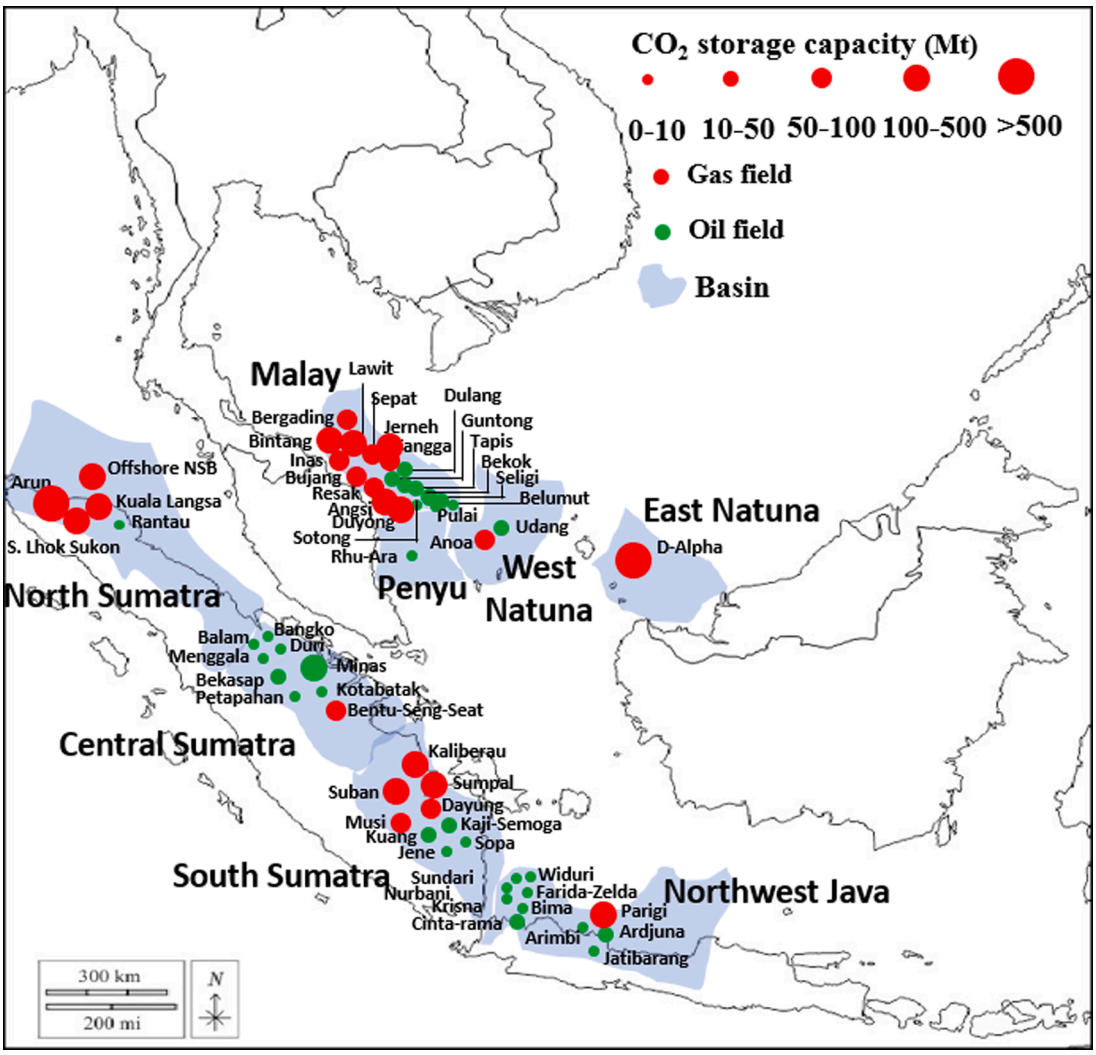


Fig. 6. Location of gas fields, oil fields, and saline aquifers within the study area.

Table 12
CO_2 storage resource in saline aquifers within the study area.

Basin	CO_2 storage (Mt)			CO_2 storage in previous studies (Mt)
	Low	Mid	High	
Central Sumatra	2,813	11,032	30,338	
Penyu	7,058	27,678	76,114	
South Sumatra	6,873	26,954	74,123	7,700 (ADB, 2013)
Malay	19,315	75,746	208,301	84,000 (Junin et al., 2016)
West Natuna	9,755	38,255	105,202	
North Sumatra	28,557	111,990	307,973	
Northwest Java	7,100	27,842	76,564	7,100 (Hedriana, 2017)
East Natuna	15,286	59,944	164,846	
Total	96,757	379,441	1,043,461	

North Sumatra, Central Sumatra and South Sumatra, NW Java basins to store several hundreds of years of CO_2 emission from within each region (Table 13). West Natuna is still under primary depletion while the East Natuna is not yet developed. Consequently, they are not ready for CO_2 storage. CO_2 emission from Peninsular Malaysia can be stored in subsurface reservoirs in the Malay and Penyu basins.

CO_2 emission from Singapore can be stored either in the Central Sumatra, North Sumatra, or Malay basins (Table 13). Central Sumatra is only 200 km from Singapore and the Minas oilfield is a good candidate for CO_2-EOR and CO_2 storage. Furthermore, Singapore is connected to the Minas oil field through the 468-km Grissik-Batam-Singapore pipeline (SPC, 2021). In addition, supply of natural gas from Sumatra to Singapore will cease in 2023 as the current contract ends. Therefore, this pipeline may be considered for CO_2 shipment from Singapore to Sumatra. The Minas oil field has a mid CO_2-EOR potential of 767 MMbbl and a mid CO_2 storage resource of 112.4 Mt (Table 11). Besides Minas, the depleted Arun gas condensate field in North Sumatra is also a good candidate for subsurface storage of CO_2 from Singapore. It has a mid CO_2

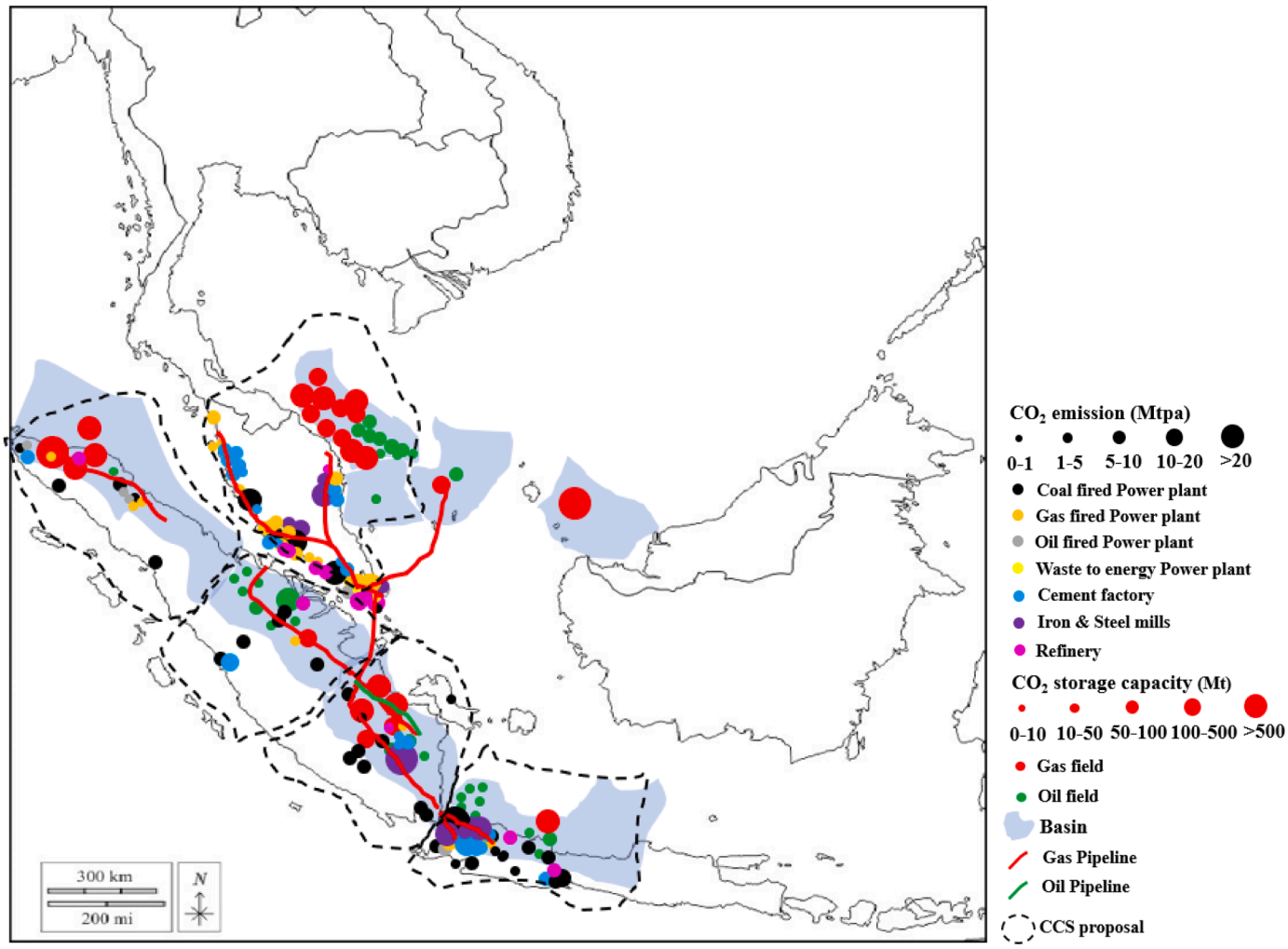


Fig. 7. Map of CO_2 sources and sinks within the study area.

storage resource of 1.3 Gt and mid CO_2-EGR condensate recovery of 101 MMbbl (Table 8). However, the Arun gas field is 800 km from Singapore. Transport of CO_2 from Singapore to Arun will probably require using liquid CO_2 tankers. The third choice for Singapore's CO_2 storage is the oil fields in the Malay Basin which is 400 km from Singapore. Shipment of CO_2 from Singapore to the Malay Basin may utilize existing natural gas pipelines or by liquid CO_2 tankers. Many of the oilfields in Malay Basin can benefit from CO_2-EOR. However, implementation of offshore CO_2-EOR will present more challenges than onshore CO_2-EOR.

Figs. 11-13 give the ranking of oil and gas fields within the study area according to EOR/EGR potential and CO_2 storage . The top-ranked oil reservoir in CO_2-EOR is Minas in Central Sumatra. Next are the Seligi, Dulang, Tapis and Guntong oil fields in the Malay Basin. The top-ranked gas reservoirs are Arun in North Sumatra and Suban in South Sumatra (Fig. 13). However, CO_2 storage in Suban will await further gas depletion. The D-Alpha gas field is big but not yet developed (Table 14).

It is worthwhile to mention that saline aquifers constitute 98% of the overall storage resource. However, CO_2 storage in them is not profitable for companies unless there is a substantial carbon tax. As of 2022, Singapore is the only Southeast Asian country with a carbon tax of S$5/tonne CO_2e (US $3.7/tonne) (MSE, 2021). This is not adequate to make geological storage of CO_2 profitable although there are plans to increase the carbon tax to S$25/tonne (US $18.3/tonne) in 2024. To make CO_2 storage in saline aquifers economically feasible, a carbon tax of at least US $50/tonne CO_2e will be needed, which is similar to the 45Q federal tax credit in the U.S. (IRS, 2021). Furthermore, saline aquifers need further geological and reservoir characterization to determine the integrity of the reservoir seals, net-to-gross, reservoir boundaries, faults, fracture gradients, etc.

4. Discussion

We propose an ASEAN CCS corridor where industrial CO_2 from Singapore is shipped to one or more of three locations for subsurface storage: (1) Minas oil field in Central Sumatra, (2) Arun gas condensate field in North Sumatra, and (3) oil and gas fields in the Malay Basin

Table 13
CO_2 sources and sinks in the study area.

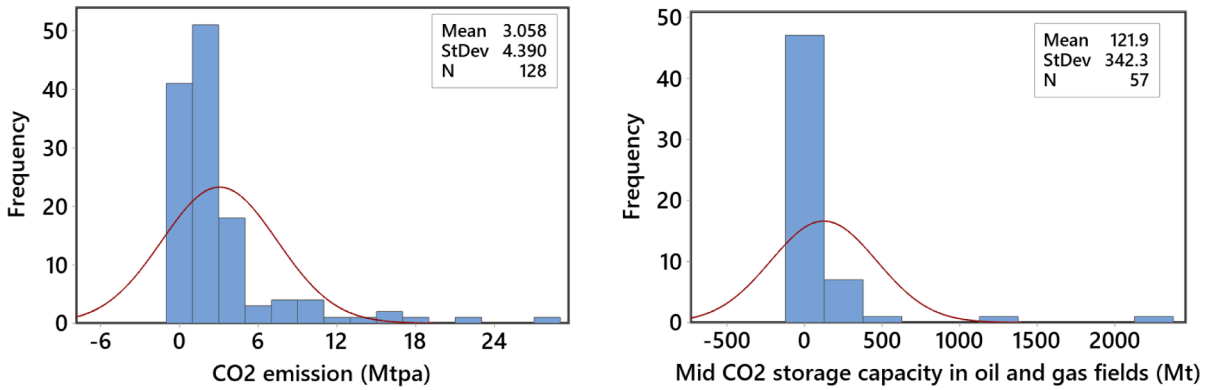
CCS clusters	CO_2 source Type of CO_2 source	Numbers of plants	CO_2 emission (Mtpa)	CO_2 sinks Type of CO_2 sinks	Numbers of fields	CO_2 storage (Mt)
CO_2 source: Central Sumatra; CO_2 sink: Central Sumatra Basin	Power plant	7	9.65	Gas	1	68
	Cement factory	1	7.47	Oil	8	160
	Refinery	1	1.71	Saline aquifer	1	11,032
	Subtotal	9	18.83	Subtotal	10	11,260
CO_2 source: South Sumatra CO_2 sink: South Sumatra Basin	Power plant	10	10.39	Gas	5	599
	Cement factory	3	1.8	Oil	4	48.4
	Refinery	1	0.71	Saline aquifer	1	26,954
	Iron & Steel mills	1	28.5			
	Subtotal	15	41.4	Subtotal	10	27,601
CO_2 source: Peninsular Malaysia CO_2 sink: Malay Basin, Penyu Basin	Power plant	17	74.74	Gas	11	1381
	Cement factory	13	23.36	Oil	9	194.8
	Refinery	4	31.33	Saline aquifer	2	103,424
	Iron & Steel mills	5	7.11			
	Subtotal	39	136.54	Subtotal	22	105,000
CO_2 source: North Sumatra; CO_2 sink: North Sumatra Basin	Power plant	11	7.2	Gas	4	1,792
	Cement factory	1	1.26	Oil	1	7.5
	Refinery	1	1.92	Saline aquifer	1	111,990
	Subtotal	13	10.38	Subtotal	6	113,790
CO_2 source: NW Java; CO_2 sink: NW Java Basin	Power plant	27	79.32	Gas	1	138
	Cement factory	5	24.12	Oil	10	66.2
	Refinery	2	6.73	Saline aquifer	1	27,842
	Iron & Steel mills	4	31.07			
	Subtotal	38	141.24	Subtotal	12	28,046
CO_2 source: Singapore	Power	10	20.08			
	Steel & iron	1	3.8			
	Refinery	3	19.17			
	Subtotal	14	43.05			
CO_2 sink: West & East Natuna				Gas	2	2238
				Oil	1	22.8
				Saline aquifer	2	98199
				Subtotal	5	100,460
	Grand Total		**391**	**Grand Total**		**386,157**

(Fig. 14). Besides CO_2 captured from Singapore, these storage sites can also accept and store industrial CO_2 from domestic sources.

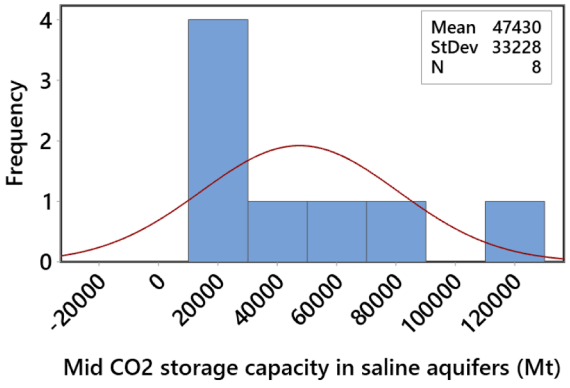
The direct distance from Singapore to Minas oil field is only 200 km. CO_2 captured from Singapore can be transported to the Minas oil field by ship, by a new pipeline, or by existing natural gas pipelines from Singapore to Grissik gas field in South Sumatra and then from Grissik to Minas. CO_2 shipment from Singapore to the Arun gas condensate field will be by liquid CO_2 tankers. CO_2 shipment from Singapore to oil and gas fields in the Malay Basin can be by liquid CO_2 tankers, or by existing natural gas pipelines, if they are available. This first-of-a-kind CCS corridor in Southeast Asia will make use of economies of scale to reduce the cost of CO_2 capture from multiple sources and subsequent shipment and injection to one or more sinks. For example, we proposed a centralized post-combustion carbon capture plant to capture CO_2 from multiple power plants and refineries in Jurong Island in Singapore (Lau et al. 2021b).

There are multiple advantages for the ASEAN CCS corridor. First, by involving more than one country, such a project can accelerate the implementation of CCS by making use of economies of scale to reduce cost in carbon capture, transportation, and storage. Second, a large-scale international CCS project can attract financing from multiple sources such as sovereign funds, venture capitalists, and international funding agencies such as the Asian Development Bank. Third, government involvement will help de-risk the long-term liability of the project by transferring long-term risk from the operator to the state. Fourth, involvement of participating governments can accelerate the promulgation of international laws for cross-border movement and storage of CO_2. Fifth, even for countries which do not participate in cross-border capturing, transporting and storage of CO_2, a system of carbon credit may be established within ASEAN countries for carbon trading. Sixth, an international project will raise the awareness and acceptance of CCS and make it easier for individual ASEAN countries to implement CCS projects within their own border. Indeed, the urgency to achieve net-zero for the region by 2050 or shortly afterwards warrants the cooperation of multiple countries as decarbonization on a regional level will be more effective than on a national level. The more Southeast Asian countries work together to solve a common problem, the more likely they will succeed by pulling the required expertise together as it is expensive and time consuming to muster such expertise on a country-by-country basis.

In recently published articles, we have proposed the implementation of a "Southern Lights Project" wherein 1 Mtpa of CO_2 from Singapore is transported to a nearby country for permanent storage in a subsurface reservoir (Lau et al., 2021a; Lau et al., 2021b). In addition to CO_2 from Singapore, the project can also store CO_2 from the host country. Indeed, such a project can be the vehicle to implement the first-of-a-kind ASEAN CCS Corridor. The Minas oil field is the biggest oil field in SE Asia (Salem, 1994) and the Arun gas field is the biggest gas condensate field in the ASEAN (Von der Mehden and Lewis, 2004). Both are suitable for CO_2 storage. The CCS corridor proposed in this study takes advantage of carbon tax in Singapore and utilizes the oil and gas field with largest



(a) CO_2 emission from stationary plants (b) Mid CO_2 storage in gas and oil fields



(c) Mid CO_2 storage resource in saline aquifers

Fig. 8. Histograms of CO_2 emission and CO_2 storage resource in the study area.

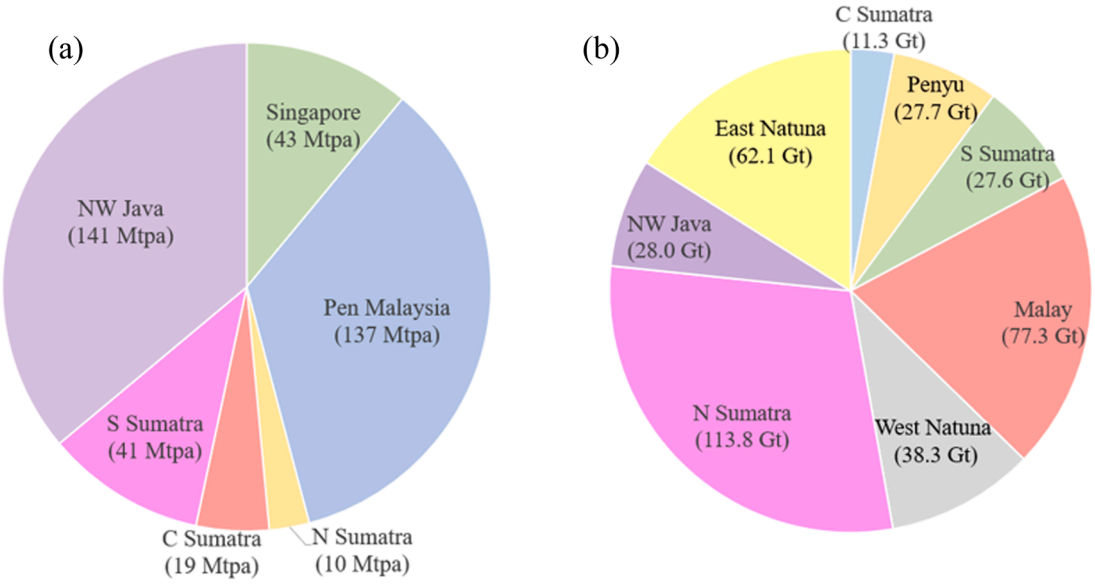


Fig. 9. (a) CO_2 emission and (b) CO_2 storage resource in the study area.

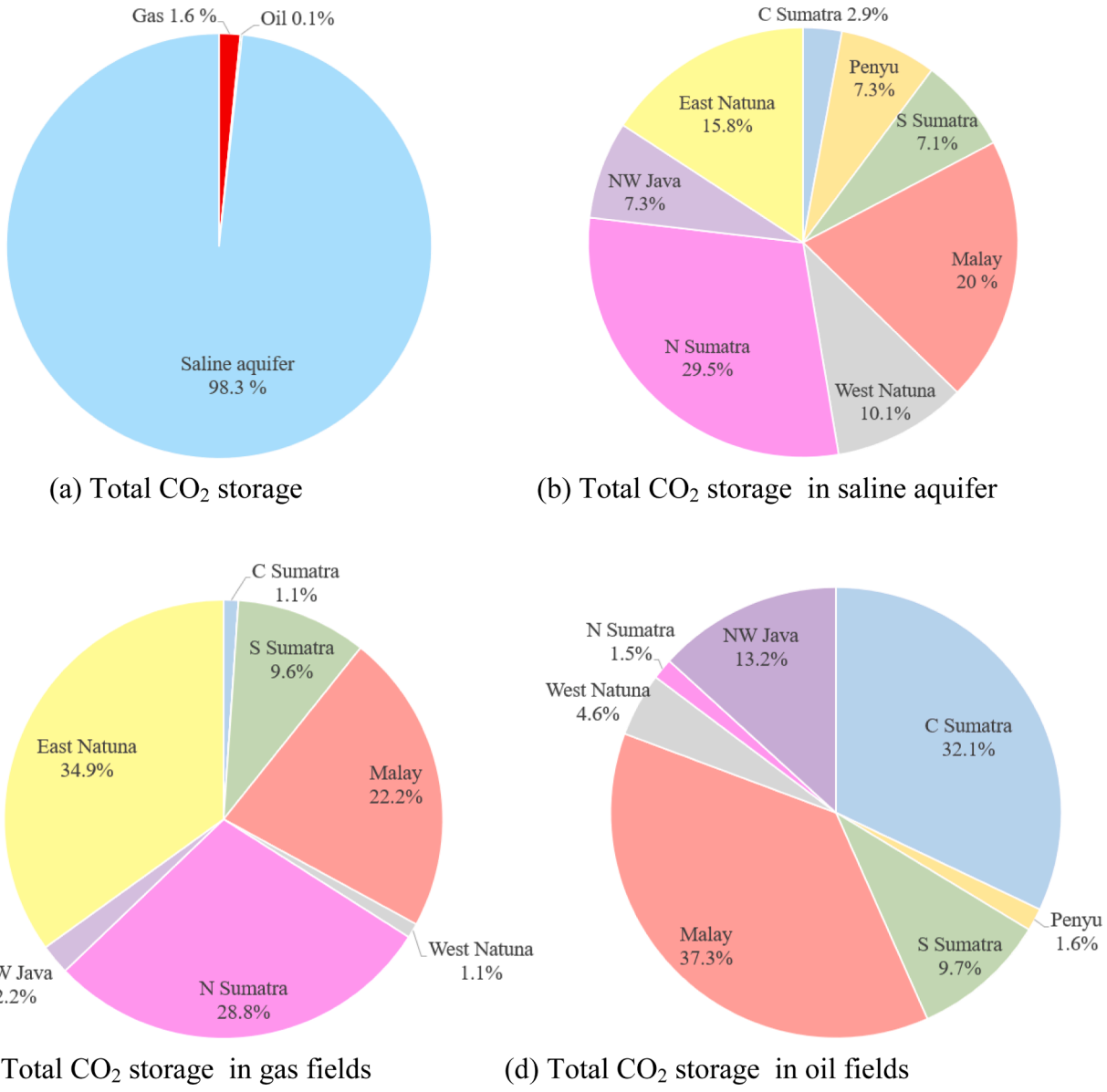


Fig. 10. Classification of CO_2 storage resource in study area by type and location.

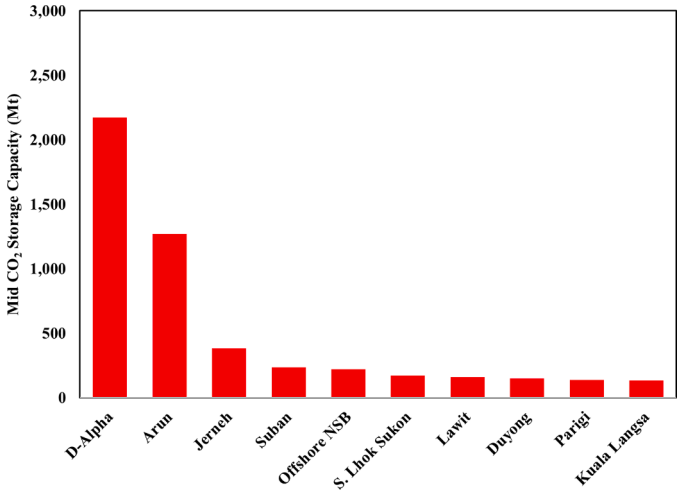


Fig. 11. Mid CO_2 storage by field within the study area

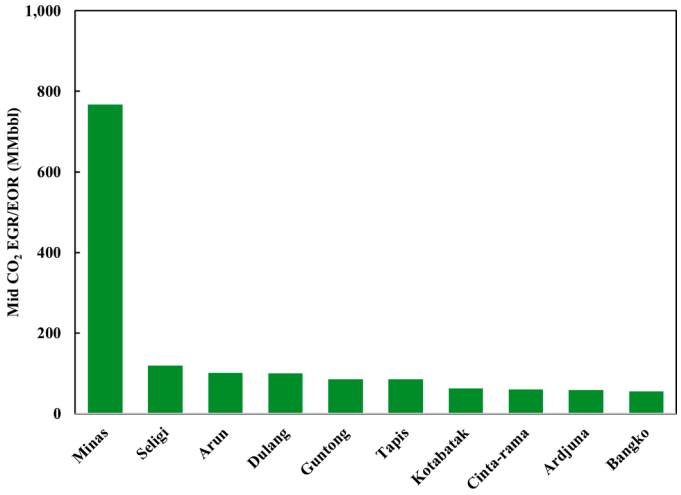


Fig. 12. Top ten CO_2-EGR/EOR by field within the study area

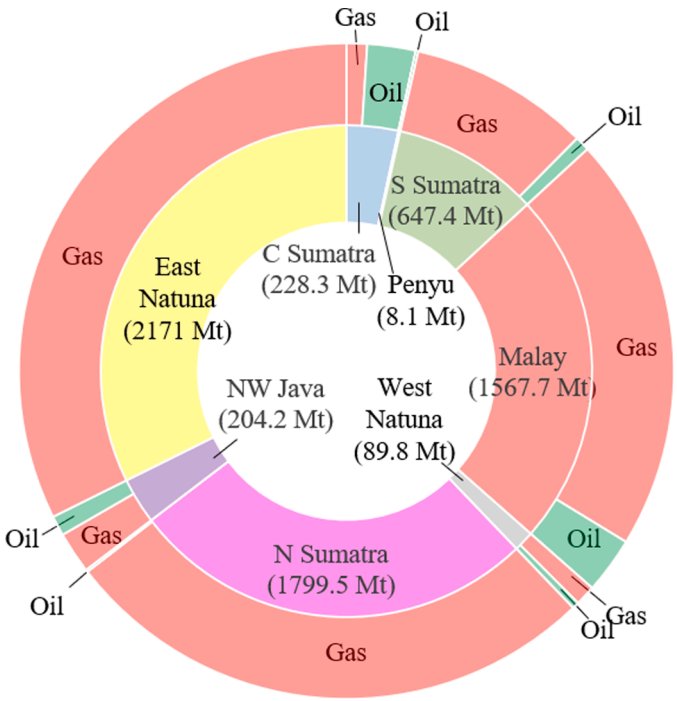


Fig. 13. CO_2 storage in oil and gas fields by basin within the study area

CO_2-EOR/EGR potential in ASEAN for CO_2 storage.

Although CCS has a high technical readiness level to mitigate CO_2 emission, there are a number of barriers to CCS projects which have been discussed by Lau et al. (2021a). For CCS projects that cross national borders, international financing will be needed (Rassool and Havercroft, 2021). For CO_2-EOR or CO_2-EGR projects, monetization of incremental oil production is one way to finance these projects. However, for CCS projects to grow fast numerically and geographically, they have to include CO_2 storage in saline aquifers (Herzog, 2017). National governments can help by imposing a credible carbon tax or a tax credit to incentivize companies to implement these projects. In addition, an international carbon trading system will also play a pivotal role in providing financing.

The introduction of a CCS corridor will use of economies of scale to reduce the overall cost of CO_2 capture, transportation and injection. Governments may provide partial funding to develop shared transport and storage networks (Rassool and Havercroft, 2021).

In addition, long-term storage liability needs to be addressed (Rassool and Havercroft, 2021). A CCS project need to include monitoring and field surveillance to detect leakage post CO_2 injection. Long term

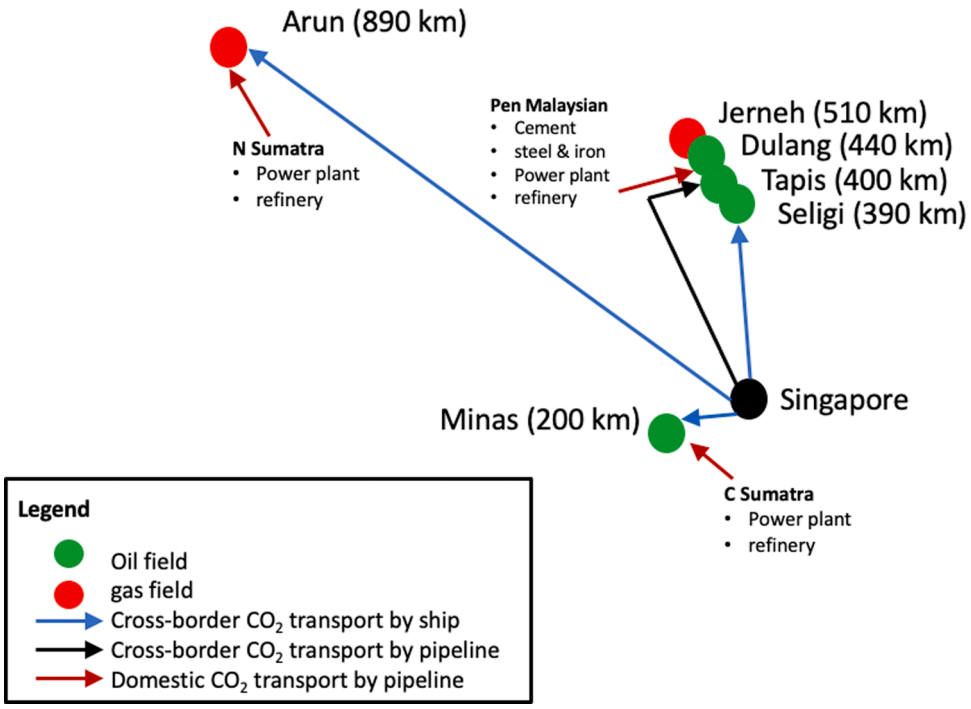


Fig. 14. An ASEAN CCS Corridor.

storage liability needs to be transferred to the state after a specified period of time.

5. Conclusions

The followings can be concluded from our study.

1 The total stationary CO_2 emission from power plants, steel and iron mills, cement factories and refineries within 1,000 km from Singapore is 391 Mtpa (Table 5). Of this, 51% comes from power plants, 24% from iron and steel mills, 15% from cement factories and 10% from refineries.
2 There is 386 Gt of mid CO_2 storage resource within 1,000 km from Singapore which is enough to store 987 years of CO_2 emission from stationary sources (Table 13). Of this, 6.2 Gt is from gas reservoirs (Table 6), 0.5 Gt from oil reservoirs (Table 9), and 379 Gt from saline aquifers which account for 98% of CO_2 storage resource (Table 12).
3 There is CO_2-EOR potential of 1,857 MMbbl in oil reservoirs within the study area (Table 9). Oil fields with the largest CO_2-EOR potential are the Minas oilfield in Central Sumatra Basin (Table 11) and the Seligi and Dulang oil fields in the Malay Basin (Table 10).
4 There is another 163 MMbbl of condensate recovery potential from CO_2-EGR within the study area (Table 6). The Arun gas condensate field in the North Sumatra Basin has the largest potential (Table 8).
5 Oil fields with light crude in the study area are ready for CO_2-EOR. However, many gas fields are still under primary depletion and not ready for CO_2 storage, except for Arun gas condensate reservoir in the North Sumatra Basin (Table 14) and Jerneh gas reservoir in the Malay Basin (Table 15).

6. Recommendation

Based on our findings, we recommend establishing an ASEAN CCS corridor connecting Singapore with the North Sumatra, Central Sumatra, and Malay basins wherein CO_2 from Singapore and the host country can be transported to oil and gas reservoirs or saline aquifers within these basins for permanent storage. This first-of-a-kind cross border CCS project has the potential to accelerate large-scale CCS implementation within ASEAN where multiple countries can benefit from economies of scale, international financing, promulgating of CCS regulations, technology transfer and rapid decarbonization of the region.

Author statement

Prof Hon Chung Lau is responsible for conceptualization. Dr. Kai Zhang is responsible for detailed analysis.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A

The properties of gas fields in the study area are given in Tables 14-15.

The properties of oil fields in the study area are given in Tables 16 and 17.

The properties of saline aquifers in the study area are given in Table 18.

Table 14
Properties of Indonesian gas reservoirs within the study area.

Basin	Gas field	Water depth (m)	Buried depth (m)	IGIP (Tcf)	ICIP (MMbbl)	Pressure (MPa)	Temperature (oC)	CO_2 density (kg/m^3)	Recovery (%)	CO_2 (%)	C_1-C_{7+} (%)	B_g	Reference
Central Sumatra	Bentu		762	1.1		8	46	236	12	0.5	99.5	0.012	(Yuwono et al., 2012; EMP, 2019)
South Sumatra	Suban		2,750	5.7	333	30	154	483	50	5.0	95.0	0.004	(Mohede et al., 2014; Ginanjar et al.,2018)
	Kaliberau		2,430	2		25	119	511	0	32.0	68.0	0.005	(Michael et al., 2007; Wiratmo et al., 2020; Koning et al., 2021)
	Dayung		1,826	1.4		17	89	449	36	30.7	69.3	0.006	(Sagita et al., 2008; Achiat et al., 2009; Suyoto et al., 2010; Abadi et al., 2016)
	Sumpal		1,696	1.6		18	83	520	30	36.0	64.0	0.006	(Suklis and Arnold Ames, 2004; Satyana et al., 2007; Darmadi et al., 2014; Permana et al., 2018; Tejo et al., 2020)
West Natuna	Musi		837	1.21		8.8	61	223	25	2.2	97.8	0.012	(Martadinata, 1999; Budi, 2001; Widarsono, 2006)
	Anoa	75	1,250	1.2	92	12.8	85	317	29	45.0	55.0	0.008	(Fahman et al., 1991; Suarsana et al., 2009; Guniawati et al., 2010; Premieroil, 2010)
North Sumatra	Arun		3,048	13.1	840	50	177	628	87	13.8	86.2	0.006	(Soeparjadi, 1983; Pathak et al., 2004)
	Kuala Langsa		3,299	3		33	148	539	0	81.0	19.0	0.004	(Caughey and Wahyudi, 1993; Sitawati et al., 2013)
	South Lhok Sukon		2,600	3		26	117	536	75	14.0	86.0	0.005	(Rory, 1990; Meckel, 2012)
	Offshore NSB	106	1,160	2.7		12	57	474	34	17.5	82.5	0.008	(Mundt, 1982; Davies,1985; Sulitra, 1991; Meckel et al., 2012)
NW Java	Parigi	50	813	2		8	40.65	269	34	1.3	98.7	0.012	(Yaman et al., 1991; Sudarmono et al., 2003; Tjahyadi et al., 2004; Yulianus et al., 2018)
East Natuna	D-Alpha	500	3,900	46.3		39.4	171	552	0	70.0	30.0	0.004	(Fenter and Hadiatno, 1996; Suhartanto et al., 2001)
	Subtotal			84.3	1,264								

Table 15
Properties of gas reservoirs in the Malay Basin.

Gasfield	Water depth (m)	Buried depth (m)	OGIP (Tcf)	OCIP (MMbbl)	Pressure (MPa)	Temperature (∘C)	CO_2 density (kg/m^3)	Recovery (%)	CO_2 (%)	C_1-C_{7+} (%)	B_g	Reference
Duyong	77	1650	2.3		17	91	438	64	3.6	96.4	0.007	(Seong et al., 1995; Seong et al., 1997; Mohammad et al., 2000; Abd Rahman, 2012)
Jerneh	55	1500	5.6		15	82.5	408	80	4.2	95.8	0.007	(rThambydurai et al., 1988; Perreau, 1993)
Lawit	57	1200	2.3		12	66	374	73	11.0	89.0	0.009	(Dong et al., 1998; Mohd et al., 2008; Osman et al., 2012)
Resak	70	2326	1.6		24.9	140	443	21	17.0	83.0	0.006	(Ong et al., 2000; Ueda et al., 2003; Omar et al., 2006)
Bintang	60	2005	2		20	110	437	50	0.1	99.9	0.006	(JICA, 1978; Madon et al., 2004)
Tangga	70	1450	1.4		15	79.75	427	0	37.0	63.0	0.007	(Bate et al., 2004; Muhamad et al., 2008; Teh et al., 2008; Sukor et al., 2020;)
Bujang	65	1400	1.5		14	77	402	0	66.0	34.0	0.006	(JICA, 1978; Mohamad et al., 2003; Madon et al., 2004; Nakayama et al., 2006)
Inas	65	1500	1		15	82.5	408	0	60.0	40.0	0.006	(JICA, 1978; Mohamad et al., 2003; Madon et al., 2004; Nakayama et al., 2006)
Sepat	65	1781	1.2		18	99	430	0	56.0	44.0	0.006	(JICA, 1978; Mohamad et al., 2003; Osman et al., 2012)
Angsi	69	2550	1.9	95	24	119	492	22	1.9	98.1	0.006	(Gopalakrishnan et al., 1999; Murrey et al., 2003; Othman et al., 2007; Mohamad et al., 2008; Abdul Hamid, 2012; Mandal et al., 2013)
Bergading	60	1219	1.4		13	66	437	28	40.0	60.0	0.007	(JICA, 1978; Jones et al., 2014)
Subtotal			22	95								

Table 16
Properties of Malaysian oil reservoirs within the study area.

Basin	Oilfield	Water depth (m)	Buried depth (m)	OOIP (MMbbl)	Pressure (MPa)	Temperature (∘C)	Recovery (%)	Bo	API	CO_2 density (kg/m^3)	Reference
Penyu	Rhu-Ara	75	2544	232	26	143	36	1.1	35	451	(Haribowo et al., 2013; Madon et al., 2019; Wilkinson et al., 2019)
Malay	Guntong	65	1600	1000	17	100	36	1.3	46	395	(Hui, 1988; Wan-Nawawi et al., 2002; Kueh et al., 2005; Selamat et al., 2011)
	Seligi	70	1275	1400	13	72	35	1.3	46	386	(Yong and Soffian, 1992; Razak et al., 2010; Kadir et al., 2021)
	Tapis	64	1800	1000	18	121	41	1.3	45	351	(Thu et al., 1983; Abdullah et al., 1999; Wan-Mohamad et al., 2005)
	Pulai	74	1205	129	12	88	37	1.3	45	270	(Cheong, 1979; Flores, 1982)
	Bekok	70	1676	476	17	94	40	1.3	47	422	(JICA, 1978; Yassin, 1988)
	Sotong	75	2165	224	22	130	30	1.3	43	415	(Md Zin Che et al, 2000a; Md Zin Che et al., 2000b)
	Belumut	72	1120	244	11	79	10	1.2	37	260	(Ong et al., 2011; Mohd Ismail et al., 2018)
	Dulang	76	1580	1167	15	99	21	1.2	38	336	(Kechut et al., 2001; Hamdan et al., 2005; Samsudin et al., 2005; Ibrahim et al., 2011)
Total	Subtotal			5871							

Table 17
Properties of Indonesian oil reservoirs within the study area.

Basin	Oilfield	Water depth (m)	Buried depth (m)	OOIP (MMbbl)	Pressure (MPa)	Temperature (∘C)	Recovery (%)	Bo	API	CO_2 density (kg/m^3)	Reference
Central Sumatra	Minas		610	9023	6	93	51	1.2	36	111	(Hasan and Kamal, 1978; Salem, 1994; Rachmawati et al., 1997; Bou-Mikael et al., 2000; Putra, 2001; Hendih et al., 2002)
	Duri		228.6	3483	2	38	30	1	21	37	(Gael et al., 1995; Kirana et al., 1997)
	Bangko		518	647	5	38	33	1.1	33	125	(Yarmanto and Indrawardana, 1995; Permadi et al., 2004; Kusumawatie et al., 2014; Yunaz et al., 2015)
	Bekasap		995	633	10	104	50	1.2	37	176	(Rahardja and Pristiwanto, 1994; Pramudyo et al., 2007)
	Menggala		1106	160	10	113	35	1.2	35	174	(Syaefuddin et al., 2001; Pederson et al., 2002)
	Balam		642	280	7	84	13	1.1	34	121	(Malone et al., 1994; Waren et al., 2015)
	Petapahan		1249	130	12	112	63	1.1	33	217	(Roezin, 1974; Anggoro et al., 2009)
	Kotabatak		1455	732	14	127	26	1.1	33	249	(Kodl, 1994; Wibowo et al., 1996; Wijaya et al., 2010)
South Sumatra	Kuang		1450	626	15	98	53	1.1	31	346	(Zeliff et al., 1985; Sardjito et al., 1991; Tamtomo et al., 1997)
	Kaji-Semoga		870	400	9	42.63	40	1.2	38	377	(Forrest, 2005; Bee, 2013)
	Sopa		2195	100	23	107.555	19	1.2	36	515	(Forrest, 2005; Bee, 2013)
	Jene		1780	150	19	87.22	37	1.2	36	525	(Forrest, 2005; Bee, 2013)
West Natuna	Udang	76	2515	330	25	116	61	1.2	39	522	(Sutoto, 1991; Abadi et al., 2012)
North Sumatra	Rantau		700	300	7	32	39	1.3	51	242	(Inoue and Ananto, 1987; Simamora et al., 2011; Suarsana et al., 2012)
NW Java	Cinta-rama	27	975	697	9.6	86	25	1.1	30	198	(Kovacs, 1982; Bishop, 2000; Forrest, 2005; Afi et al., 2015)
	Krisna	21	1433	120	14.2	99	43	1.2	36	310	(Bishop, 2000; Forrest, 2005)
	Widuri	21	1076	260	11	53.8	27	1.1	32	430	(Elzinga et al., 1991; Wight et al., 1997; Carter et al., 1998; Aji et al., 2013; Widiatmo et al., 2017)
	Bima	27	777	700	7.9	67	14	1	24	175	(Prior, 1986; Crumb, 1989; Woodling et al., 1990)
	Farida-Zelda	21	2134	216	21.1	124	50	1.1	32	414	(Stewart and Sitinjak, 1989; Sumii et al., 1992; Kim et al., 1996; Noble et al., 1997)
	Sundari	24	1326	100	13.5	110	24	1	22	263	(Wight et al., 1997; Ginanjar et al., 2014)
	Nurbani	24	518	100	5.3	60	0	1	15	106	(Wight et al., 1997; Aji et al., 2012)
	Ardjuna	23	725	683	7	40	50	1.2	37	198	(Davidson and Victors, 1986; Sumii et al., 1992; Paskvan et al., 1996; Kim et al., 1996; Noble et al., 1997)
	Arimbi	30	741	116	7.5	79	53	1.1	33	148	(Caycedo and Diebold, 1979; Sumii et al., 1992; Kim et al., 1996; Noble et al., 1997)
	Jatibarang		2282	250	21.9	119	36	1.1	30	450	(Gunadi et al., 2005; Suarsana et al., 2005; Sahudi et al., 2005)
Total	Subtotal			26,107							

Table 18
Properties of saline aquifers within the study area.

Basin	Area (km^2)	Net sand thickness (m)	Average pressure (MPa)	Average temperature (°C)	Average CO_2 density (kg/m^3)	Porosity (%)	Reference
Central Sumatra	117,300	150	13	138	209	15	(APEC, 2005)
Penyu	51,000	450	15	84	402	15	(APEC, 2005)
South Sumatra	133,700	300	12	110	224	15	(APEC, 2005)
Malay	83,000	450	15	70	507	20	(APEC, 2005; Sihombing et al., 2019)
West Natuna	69,454	450	15	83	408	15	(APEC, 2005; Cherdasa et al., 2018)
North Sumatra	258,000	390	15	90	371	15	(Hasbollah et al., 2020)
NW Java	140,870	270	12	99	244	15	(Fanani et al., 2006; Madon et al., 2019)
East Natuna	71,927	450	15	75	463	20	(Wight and Reynolds, 1995)

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